

Diffusion-Driven Graph Optimization Framework for Bundle Recommendation

Anonymous submission

Abstract

Existing bundle recommendation methods typically use multi-view information from user–bundle, user–item, and bundle–item graphs to accurately model users’ bundle preferences. However, such multi-view methods still face two key challenges when learning from these graphs: (1) unreliable user–bundle interactions may propagate misleading signals and impair preference learning; (2) the sparsity of user–bundle implicit feedback limits the available supervision. We therefore propose a diffusion-driven graph optimization framework for bundle recommendation (DiGO). First, we aggregate pretrained representations of the bundles in each user’s interaction history and condition a diffusion process on the pretrained user representation to model the user’s overall bundle preference. We then use this preference representation to score observed user–bundle interactions, retain more reliable user–bundle interactions, and reconstruct the user–bundle graph for subsequent multi-view representation learning. Second, to address the sparsity of user–bundle implicit feedback, we use representations learned from the reconstructed user–bundle graph as anchors and align user–item and bundle–item representations through cross-view contrastive learning before fusing the three views. This allows the auxiliary relations to supplement the limited supervision from user–bundle feedback. Experiments on three public datasets show that our framework outperforms state-of-the-art methods on all evaluation metrics, achieving a maximum relative improvement of 10% over the strongest baseline.

1 Introduction

Bundle recommendation aims to identify bundles that match users’ interests from a set of predefined item collections. Existing methods typically model user–bundle, user–item, and bundle–item relations jointly and learn complementary user and bundle representations through graph neural networks, multi-view fusion, and contrastive learning (????). Among these relations, user–item and bundle–item relations provide auxiliary information, whereas the user–bundle graph directly carries users’ implicit feedback on bundles. Its structural and supervisory quality therefore affects subsequent

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multi-view learning. Despite the progress of existing methods, multi-view bundle recommendation still faces two key challenges.

Challenge 1: Unreliable interactions in the user–bundle graph. User–bundle relations are typically constructed from implicit feedback and may contain interactions that do not reliably reflect users’ preferences. For example, accidental clicks, exploratory choices, and interactions caused by short-term interests or exposure bias may fail to consistently represent users’ actual bundle preferences. Existing graph-based methods propagate information directly along observed edges, allowing unreliable interactions to participate in representation updates and introduce misleading signals. Therefore, identifying more reliable connections from users’ interaction histories and constructing a high-quality user–bundle graph remain key challenges.

Challenge 2: Sparse user–bundle implicit feedback. User–bundle feedback is substantially sparser than user–item interactions, providing limited bundle-level supervision. Although user–item and bundle–item relations offer rich auxiliary signals, final-stage fusion alone may not fully align them with user–bundle preference learning. Therefore, effectively leveraging auxiliary views while preserving consistency with direct bundle preferences remains challenging.

To address these challenges, we propose DiGO, a diffusion-driven graph optimization framework for bundle recommendation. DiGO first aggregates pretrained representations of each user’s historical bundles and conditions a diffusion process on the pretrained user representation to model the user’s overall bundle preference. The resulting preference representation is then used to score observed user–bundle interactions, retain more reliable connections, and reconstruct the graph without introducing new edges. Finally, DiGO uses representations from the reconstructed user–bundle graph as anchors to align the user–item and bundle–item views before fusion, thereby supplementing sparse supervision. Figure 1 illustrates the overall framework.

The main contributions of this work are summarized as follows:

- We propose DiGO, a diffusion-driven graph optimization framework that models users’ overall bundle preferences through conditional diffusion and uses them to re-evaluate observed interactions and reconstruct the user–bundle graph.

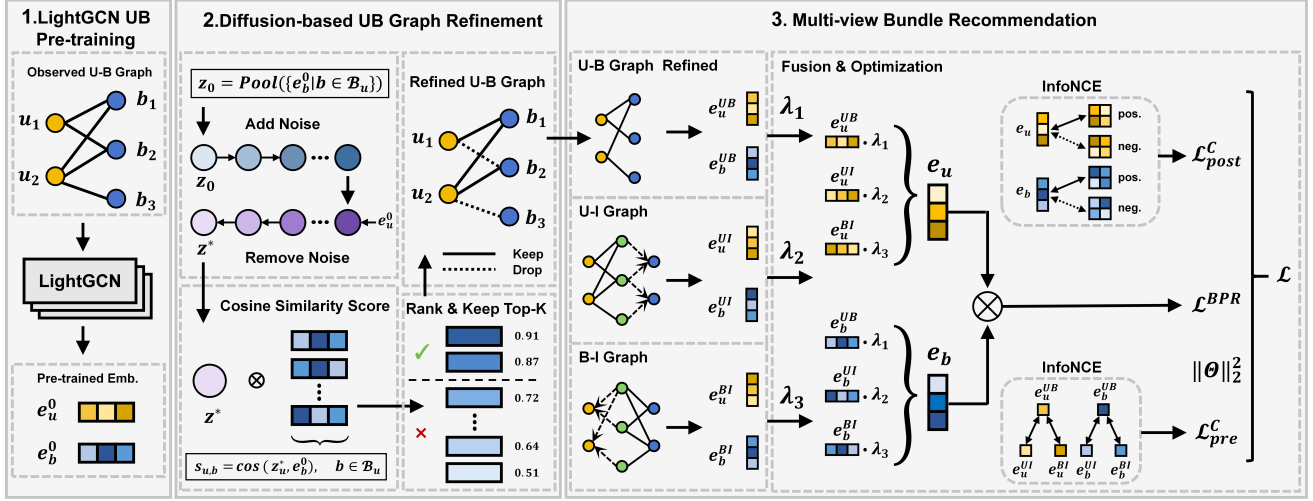


Figure 1: Overall framework of DiGO, comprising (1) user–bundle pretraining, (2) diffusion-driven graph optimization for reconstructing the observed user–bundle graph, and (3) multi-view recommendation using the reconstructed user–bundle view as the anchor for pre-fusion cross-view contrastive learning.

- We use representations learned from the reconstructed user–bundle graph as anchors to align the two auxiliary views before view fusion, thereby supplementing sparse user–bundle supervision.
- Experiments on three public datasets show that DiGO outperforms existing methods across all evaluation metrics, achieving a maximum relative improvement of 10%. Ablation and further analyses also validate the effectiveness of the two key designs.

2 Related Work

2.1 Bundle Recommendation

Early bundle recommendation methods primarily relied on user–bundle interactions and employed collaborative filtering, matrix factorization, or neural models to learn user and bundle representations (??). Subsequent graph-based methods further modeled user–bundle, user–item, and bundle–item relations to capture high-order collaborative information at both the bundle and item levels (??).

Recent studies have increasingly focused on multi-view representation learning and cross-view collaboration. For example, MIDGN introduces intent disentanglement, CrossCBR employs cross-view contrastive learning, BundleGT uses graph Transformers to model structural information, and MultiCBR learns unified representations through three-view fusion and contrastive learning (????). These methods mainly improve the encoding, fusion, or alignment of multiple views, while typically using the observed user–bundle graph directly. In contrast, DiGO re-evaluates observed user–bundle interactions before multi-view learning and reconstructs the user–bundle graph accordingly.

2.2 Diffusion Models for Recommendation

Diffusion models exhibit strong distribution modeling capabilities by progressively perturbing data and learning the cor-

responding reverse process. In recent years, they have been applied to collaborative filtering, sequential recommendation, and knowledge graph recommendation to model user preferences, recover interaction signals, or enhance recommendation representations (??).

Diffusion models have also been introduced into bundle recommendation. RDiffBR addresses dynamic bundle–item relations by applying residual diffusion to item-level bundle representations, enabling the model to adapt to changes in bundle composition (?). DCBR applies conditional diffusion to user–bundle interaction data and combines it with multi-view contrastive learning (?). Unlike these methods, DiGO aggregates the pretrained representations of bundles in each user’s interaction history and conditions the diffusion process on the pretrained user representation to model the user’s overall bundle preference. The resulting preference representation is then used to evaluate observed user–bundle interactions and reconstruct the user–bundle graph for subsequent multi-view representation learning.

3 Methodology

3.1 Problem Formulation

Let $\mathcal{U} = \{u_1, u_2, \dots, u_M\}$, $\mathcal{B} = \{b_1, b_2, \dots, b_N\}$, and $\mathcal{I} = \{i_1, i_2, \dots, i_O\}$ denote the sets of users, bundles, and items, respectively, where M , N , and O are the corresponding set sizes. We use a user–bundle interaction matrix $\mathbf{X} \in \{0, 1\}^{M \times N}$, a user–item interaction matrix $\mathbf{Y} \in \{0, 1\}^{M \times O}$, and a bundle–item affiliation matrix $\mathbf{Z} \in \{0, 1\}^{N \times O}$. Specifically, $x_{ub} = 1$ and $y_{ui} = 1$ indicate that user u has interacted with bundle b and item i , respectively, while $z_{bi} = 1$ indicates that item i belongs to bundle b .

Given $\mathbf{X}$, $\mathbf{Y}$, and $\mathbf{Z}$, the goal of bundle recommendation is to predict the preference score $y_{u,b}^*$ for each unobserved user–bundle pair.

3.2 Framework Overview

As illustrated in Figure 1, DiGO first models each user’s overall bundle preference using pretrained representations and reconstructs the original user–bundle matrix $\mathbf{X}$ as $\tilde{\mathbf{X}}$. It then learns user and bundle representations from three views based on $\tilde{\mathbf{X}}$, $\mathbf{Y}$, and $\mathbf{Z}$. Before fusing the three views, DiGO performs cross-view alignment using the user–bundle view as the anchor and finally predicts user preferences from the fused representations.

3.3 Diffusion-Driven Graph Optimization

Preference Representation Construction We first pre-train LightGCN (?) on the original user–bundle interactions to obtain pretrained representations $\bar{\mathbf{e}}_u$ and $\bar{\mathbf{e}}_b$ for user u and bundle b , respectively. The pretraining model is optimized with the BPR objective, and the resulting representations remain fixed during subsequent graph optimization.

Let $\mathcal{N}_u^{UB} = \{b \in \mathcal{B} \mid x_{ub} = 1\}$ denote the set of bundles historically interacted with by user u . We construct the historical bundle representation through mean pooling:

$$\mathbf{p}_u^0 = \frac{1}{|\mathcal{N}_u^{UB}|} \sum_{b \in \mathcal{N}_u^{UB}} \bar{\mathbf{e}}_b, \quad (1)$$

where $\mathbf{p}_u^0$ summarizes the overall characteristics of the bundles in the user’s interaction history and serves as the modeling target of the subsequent diffusion process.

Conditional Diffusion Given the number of diffusion steps T and a noise schedule $\{\beta_t\}_{t=1}^T$, we define $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$. At timestep t , the forward perturbation of $\mathbf{p}_u^0$ is defined as

$$\mathbf{p}_u^t = \sqrt{\bar{\alpha}_t} \mathbf{p}_u^0 + \sqrt{1 - \bar{\alpha}_t} \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (2)$$

We use the pretrained user representation $\bar{\mathbf{e}}_u$ as a personalized condition and employ a conditional denoising network to predict the unperturbed bundle preference representation:

$$\hat{\mathbf{p}}_u^0 = f_\theta(\mathbf{p}_u^t, \bar{\mathbf{e}}_u, t), \quad (3)$$

where f_θ is an MLP-based conditional denoising network that integrates the perturbed representation, the user condition, and the timestep embedding.

During training, we uniformly sample t from $\{1, \dots, T\}$ and draw Gaussian noise $\boldsymbol{\epsilon}$ to construct $\mathbf{p}_u^t$. The diffusion module is optimized using a representation reconstruction loss and a historical-set constraint:

$$\begin{aligned} L_{\text{rec}}^D &= \mathbb{E}_{u \sim \mathcal{U}^+, t \sim \text{Unif}\{1, \dots, T\}, \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})} \left[\|\hat{\mathbf{p}}_u^0 - \mathbf{p}_u^0\|_2^2 \right], \\ L_{\text{set}}^D &= \mathbb{E}_{u \sim \mathcal{U}^+, t \sim \text{Unif}\{1, \dots, T\}, \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})} \left[1 - \frac{1}{|\mathcal{N}_u^{UB}|} \sum_{b \in \mathcal{N}_u^{UB}} \cos(\hat{\mathbf{p}}_u^0, \bar{\mathbf{e}}_b) \right], \\ L^D &= L_{\text{rec}}^D + \eta L_{\text{set}}^D, \end{aligned} \quad (4)$$

where $\mathcal{U}^+ = \{u \in \mathcal{U} \mid |\mathcal{N}_u^{UB}| > 0\}$, $\cos(\cdot, \cdot)$ denotes cosine similarity, and η controls the contribution of

the historical-set constraint. L_{rec}^D encourages the prediction to recover the overall representation of the user’s historical bundles, while L_{set}^D encourages consistency with the actual bundles in the user’s interaction history.

Interaction Scoring and Graph Reconstruction During graph reconstruction, we employ a deterministic iterative denoising procedure. The historical bundle representation is used as the initial state, and the conditional denoising network is repeatedly applied in reverse timestep order:

$$\begin{aligned} \tilde{\mathbf{p}}_u^T &= \mathbf{p}_u^0, \\ \tilde{\mathbf{p}}_u^{t-1} &= f_\theta(\tilde{\mathbf{p}}_u^t, \bar{\mathbf{e}}_u, t), \quad t = T, T-1, \dots, 1. \\ \mathbf{p}_u^* &= \tilde{\mathbf{p}}_u^0, \end{aligned} \quad (5)$$

Here, $\mathbf{p}_u^*$ denotes the resulting overall bundle preference representation of user u .

We then compute preference-matching scores only for the user’s observed interactions and retain at most the top- κ bundles:

$$\begin{aligned} s_{u,b} &= \cos(\mathbf{p}_u^*, \bar{\mathbf{e}}_b), \quad b \in \mathcal{N}_u^{UB}, \\ \tilde{\mathcal{N}}_u^{UB} &= \text{TopK}_{b \in \mathcal{N}_u^{UB}}(s_{u,b}, \kappa), \\ \tilde{x}_{ub} &= \mathbb{I}[b \in \tilde{\mathcal{N}}_u^{UB}]. \end{aligned} \quad (6)$$

Here, κ is the maximum number of interactions retained for each user, and $\mathbb{I}[\cdot]$ is the indicator function. All $\tilde{x}_{ub}$ form the reconstructed user–bundle matrix $\tilde{\mathbf{X}}$. Since $\tilde{\mathcal{N}}_u^{UB} \subseteq \mathcal{N}_u^{UB}$, DiGO only re-evaluates and filters observed user–bundle interactions without introducing unobserved connections.

3.4 Multi-View Representation Learning

Based on $\tilde{\mathbf{X}}$, $\mathbf{Y}$, and $\mathbf{Z}$, DiGO learns representations from the user–bundle interaction view, user–item interaction view, and bundle–item affiliation view. The three views share the initial trainable user, bundle, and item representations while employing independent propagation paths.

Let $\text{LG}(\cdot)$ denote a LightGCN encoder with neighborhood propagation and layer-wise aggregation. The representation learning process for the three views is written as

$$\begin{aligned} (\mathbf{E}_U^{UB}, \mathbf{E}_B^{UB}) &= \text{LG}(\tilde{\mathbf{X}}), \\ (\mathbf{E}_U^{UI}, \mathbf{E}_I^{UI}) &= \text{LG}(\mathbf{Y}), \\ \mathbf{e}_b^{UI} &= \frac{1}{|\mathcal{N}_b^{BI}|} \sum_{i \in \mathcal{N}_b^{BI}} \mathbf{e}_i^{UI}, \\ (\mathbf{E}_B^{BI}, \mathbf{E}_I^{BI}) &= \text{LG}(\mathbf{Z}), \\ \mathbf{e}_u^{BI} &= \frac{1}{|\mathcal{N}_u^{UI}|} \sum_{i \in \mathcal{N}_u^{UI}} \mathbf{e}_i^{BI}. \end{aligned} \quad (7)$$

Here, $\mathcal{N}_b^{BI}$ denotes the set of items contained in bundle b , while $\mathcal{N}_u^{UI}$ denotes the set of items interacted with by user u . The UB view directly produces the user representation $\mathbf{e}_u^{UB}$ and bundle representation $\mathbf{e}_b^{UB}$. In the UI view, item

representations are aggregated according to bundle–item affiliations to obtain $\mathbf{e}_b^{BI}$. In the BI view, item representations are aggregated according to user–item interactions to obtain $\mathbf{e}_u^{BI}$. This three-view organization follows the standard multi-view modeling framework for bundle recommendation (?).

Consequently, user u obtains three view-specific representations $\{\mathbf{e}_u^{UB}, \mathbf{e}_u^{UI}, \mathbf{e}_u^{BI}\}$, while bundle b obtains $\{\mathbf{e}_b^{UB}, \mathbf{e}_b^{UI}, \mathbf{e}_b^{BI}\}$.

3.5 Cross-View Contrastive Learning

To exploit the two auxiliary views for supplementing sparse user–bundle supervision, DiGO uses the representations learned from the reconstructed user–bundle view as anchors and aligns them with the user–item and bundle–item representations before fusing the three views.

For an entity set $\mathcal{S}$, the contrastive loss between an anchor view a and an auxiliary view v is defined as

$$\ell_\tau(\mathbf{E}^a, \mathbf{E}^v; \mathcal{S}) = -\frac{1}{|\mathcal{S}|} \sum_{x \in \mathcal{S}} \log \frac{\exp(\cos(\mathbf{e}_x^a, \mathbf{e}_x^v)/\tau)}{\sum_{x' \in \mathcal{S}} \exp(\cos(\mathbf{e}_x^a, \mathbf{e}_{x'}^v)/\tau)}. \quad (8)$$

Representations of the same entity across the two views form a positive pair, while representations of the remaining entities serve as in-batch negatives.

The pre-fusion cross-view contrastive loss is defined as

$$L^A = \ell_{\tau_a}(\mathbf{E}_U^{UB}, \mathbf{E}_U^{UI}; \mathcal{S}_U) + \ell_{\tau_a}(\mathbf{E}_U^{UB}, \mathbf{E}_U^{BI}; \mathcal{S}_U) + \ell_{\tau_a}(\mathbf{E}_B^{UB}, \mathbf{E}_B^{UI}; \mathcal{S}_B) + \ell_{\tau_a}(\mathbf{E}_B^{UB}, \mathbf{E}_B^{BI}; \mathcal{S}_B), \quad (9)$$

where $\mathcal{S}_U$ and $\mathcal{S}_B$ denote the sets of unique users and positive bundles in the current batch, respectively, and τ_a is the temperature parameter. DiGO only constructs UB–UI and UB–BI alignment on both the user and bundle sides, without introducing an additional UI–BI constraint.

3.6 Prediction and Model Optimization

The user and bundle representations from the three views are fused using view coefficients, and their inner product is used for preference prediction:

$$\begin{aligned} \mathbf{e}_u &= \lambda_1 \mathbf{e}_u^{UB} + \lambda_2 \mathbf{e}_u^{UI} + \lambda_3 \mathbf{e}_u^{BI}, \\ \mathbf{e}_b &= \lambda_1 \mathbf{e}_b^{UB} + \lambda_2 \mathbf{e}_b^{UI} + \lambda_3 \mathbf{e}_b^{BI}, \\ y_{u,b}^* &= \mathbf{e}_u^\top \mathbf{e}_b, \quad \lambda_1 + \lambda_2 + \lambda_3 = 1. \end{aligned} \quad (10)$$

DiGO also retains post-fusion self-supervised contrastive learning. Let $\mathbf{E}'_U, \mathbf{E}''_U$ and $\mathbf{E}'_B, \mathbf{E}''_B$ denote the user and bundle representations generated by two independent representation perturbations, respectively. The post-fusion contrastive loss is

$$L^C = \frac{1}{2} [\ell_{\tau_c}(\mathbf{E}'_U, \mathbf{E}''_U; \mathcal{S}_U) + \ell_{\tau_c}(\mathbf{E}'_B, \mathbf{E}''_B; \mathcal{S}_B)], \quad (11)$$

where τ_c is the temperature parameter for post-fusion contrastive learning.

For each training triplet (u, b, b') , where $x_{ub} = 1$ and $x_{ub'} = 0$, the recommendation loss and the overall objective of the multi-view recommendation model are defined as

$$\begin{aligned} L^{BPR} &= \sum_{(u,b,b')} -\ln \sigma(y_{u,b}^* - y_{u,b'}^*), \\ L^R &= L^{BPR} + \beta_1 L^C + \beta_2 L^A + \beta_3 \|\Theta\|_2^2, \end{aligned} \quad (12)$$

where Θ denotes the trainable parameters of the multi-view recommendation model, and β_1, β_2 , and β_3 control the contributions of the post-fusion contrastive loss, the pre-fusion cross-view contrastive loss, and the regularization term, respectively.

During training, DiGO first optimizes the diffusion module with L^D and reconstructs $\tilde{\mathbf{X}}$. It then uses $\tilde{\mathbf{X}}, \mathbf{Y}$, and $\mathbf{Z}$ as inputs and optimizes the multi-view recommendation model with L^R .

3.7 Complexity Analysis

Let d, K , and Q denote the embedding dimension, the number of graph propagation layers, and the number of recommendation mini-batches per epoch, respectively. Let C_f denote the cost of one forward pass through the conditional denoising network. For the fixed-depth MLP used in DiGO, $C_f = \mathcal{O}(d^2)$ when its hidden and timestep embedding dimensions scale with d .

Let $\mathcal{E}_{UB}, \mathcal{E}_{UI}$, and $\mathcal{E}_{BI}$ denote the edge sets of the original user–bundle, user–item, and bundle–item graphs, respectively, and let $\tilde{\mathcal{E}}_{UB}$ denote the edge set of the reconstructed user–bundle graph. The offline LightGCN pre-training incurs $\mathcal{O}(K|\mathcal{E}_{UB}|d)$ for graph propagation. During each DiGO training epoch, diffusion learning costs $\mathcal{O}(|\mathcal{U}^+|C_f + |\mathcal{E}_{UB}|d)$, where the second term arises from historical bundle aggregation and the historical-set constraint.

Graph reconstruction further requires $\mathcal{O}(T|\mathcal{U}^+|C_f + |\mathcal{E}_{UB}|d)$ for iterative denoising and observed-interaction scoring, together with at most $\mathcal{O}(|\mathcal{E}_{UB}|\log \kappa)$ for top- κ selection. The multi-view recommendation component costs $\mathcal{O}\left(QKd\left(|\tilde{\mathcal{E}}_{UB}| + |\mathcal{E}_{UI}| + |\mathcal{E}_{BI}|\right)\right)$ for graph propagation and $\mathcal{O}(Q(B_U^2 + B_B^2)d)$ for pre- and post-fusion contrastive learning, where B_U and B_B are the numbers of unique users and bundles in a mini-batch, respectively.

Since $|\tilde{\mathcal{E}}_{UB}| \leq |\mathcal{E}_{UB}|$, DiGO does not increase graph propagation costs through graph densification and avoids scoring all $M \times N$ user–bundle pairs. Its parameter space complexity is $\mathcal{O}((M + N + O)d + P_f)$, where P_f is the number of parameters in the conditional denoising network.